



Luis Ángel Almazán López

AI & Software Developer ·
Electromechanical Engineer

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PROFILE

AI and software engineer with 7+ years building production systems across LLMs, computer vision, full-stack web, identity, data pipelines and industrial automation. Currently developing AI-powered tooling for automotive embedded systems at Visteon and a Manufacturing Execution System for Grupo Arga. I use Python and Rust as my primary stacks and work in a Spec-Driven Development style — features start as a precise written spec that drives both AI-assisted implementation and review. My electromechanical engineering background gives me a direct bridge between software and the factory floor. M.Sc. in Computer Engineering, with a thesis on few-shot learning for audio recognition. Active lecturer at La Salle Chihuahua.

EXPERIENCE

AI & Software Developer · Visteon

Jun 2024 – Present · Chihuahua, MX

- Build LLM-powered tools (RAG, agents, tool-calling) for embedded-systems engineering workflows using LangChain, HuggingFace and Azure OpenAI.
- Develop computer-vision pipelines in PyTorch and TensorFlow for automotive quality and validation use cases.
- Ship internal React dashboards and FastAPI services; deploy and monitor models on Azure ML.
- Collaborate with multidisciplinary engineering teams to translate hardware/firmware needs into AI features.

Software Developer (MES) · Grupo Arga

Nov 2024 – Nov 2025 · Chihuahua, MX

- Design and build a Manufacturing Execution System for an electric-transformer plant — production, traceability and quality flows.
- Backend in FastAPI / Django on PostgreSQL; containerized with Docker and deployed on Kubernetes (AWS).
- Integrate plant-floor data sources and define event/state models for live production tracking.
- Set up CI/CD, Linux-based infrastructure and observability for plant-floor reliability.

Professor · La Salle Chihuahua

Jan 2023 – Dec 2025 · Chihuahua, MX

- Lecturer for Artificial Intelligence, Discrete Mathematics, Differential Equations, Instrumentation & Control, Mechanical Design, Dynamics, Microcontrollers (PICs) and Signal Processing.
- Mentor students on real-world software development, OOP, version control and applied ML projects.
- Design hands-on assignments that connect classroom theory with industry tools.

Data Engineer & ML Engineer · Leading Tech

Jul 2023 – Apr 2024 · Chihuahua, MX

- Built large-scale web-scraping pipelines (Scrapy, Selenium) over news outlets for public-figure and brand-perception analytics.
- Implemented NLP models (NLTK, HuggingFace transformers, TensorFlow) for classification and entity extraction across thousands of documents per day.
- Designed ETL into PostgreSQL/MongoDB and served models behind Flask APIs.
- Owned end-to-end ML workflows: dataset curation, training, evaluation and deployment.

ML Engineer · MVP

Jul 2022 – Jan 2023 · Chihuahua, MX

- Built a vehicle license-plate recognition and classification system (detection + OCR + status logic) for parking-operations use cases.
- Tuned detection models and post-processing rules to operate reliably on real-world video.

Electromechanical Engineer · FPA

Jul 2021 – Jan 2023 · Chihuahua, MX

- Led electrical and automation design for industrial projects; programmed Siemens / Allen-Bradley PLCs.
- Produced electrical schematics in EPLAN and mechanical designs in SolidWorks to improve production efficiency.
- Coordinated with maintenance and production teams to commission new equipment on-site.

Electromechanical Maintenance Engineer · Interceramic

Jan 2021 – Jul 2021 · Chihuahua, MX

- Owned preventive and corrective maintenance for industrial electromechanical equipment, reducing downtime and improving line reliability.
- Diagnosed PLC and electrical faults on the production line and coordinated repairs with operators.

CNC Programmer · Semain

Aug 2020 – Jan 2021 · Chihuahua, MX

- Programmed 3-axis CNC lathes and mills using CAMWorks and BobCAM; optimized toolpaths for cycle-time and part quality.
- Generated CAD/CAM documentation from SolidWorks models for the machining team.

Full-Stack Developer · MVP

Dec 2019 – Jul 2022 · Chihuahua – Delicias, MX

- Automated parking-meter operations for the city of Chihuahua: Java backend, React frontend.
- Worked part-time (remote and on-site) in parallel with engineering studies, owning features end-to-end.

Electric Supervisor · Instalaciones Eléctricas y Automatización

Aug 2018 – Jul 2020 · Delicias, MX

- Supervised industrial electrical installations and PLC programming on large-scale projects.
- Coordinated technicians and contractors, ensuring safety standards and on-time delivery.

EDUCATION

M.Sc. Computer Engineering · Universidad Autónoma de Chihuahua (UACH)

Jan 2022 – Dec 2023

Thesis: machine learning for audio recognition using few-shot learning techniques.

B.S. Electromechanical Engineering · Instituto Tecnológico de Delicias

Aug 2016 – Dec 2020

Best GPA of the LVIII generation; one of the few to graduate in 9 semesters.

SKILLS

Languages: Python (Expert), Rust, TypeScript / JavaScript, C# / .NET, Java, C / C++

AI / Machine Learning: PyTorch, TensorFlow / Keras, HuggingFace, LangChain, RAG pipelines, Computer Vision (OpenCV, CNNs, YOLO, ArcFace), ONNX Runtime, MLflow, Label Studio, Azure ML, scikit-learn

Web & Backend: FastAPI, Flask, Django, Axum / Tokio (Rust), Express, React, REST APIs, MCP servers

Data & Streaming: PostgreSQL, MongoDB, MySQL, Oracle, SQL Server, Qdrant, ClickHouse, Redis, Kafka (Avro)

DevOps & Tools: Git / GitHub, Docker, Kubernetes (AWS EKS), Linux, Ansible, Prometheus / OpenTelemetry, Spec-Driven Development

Engineering: PLC Programming (Siemens, Allen-Bradley), EPLAN, SolidWorks, CNC / CAM (CAMWorks, BobCAM)

Soft skills: Leadership, teaching, technical communication, cross-functional collaboration

Languages (spoken): Spanish (native), English (professional), French (basic)

SELECTED PROJECTS

- **Multi-tenant Enterprise AI Assistant** — Chat platform with multi-provider LLM routing and a modular MCP server architecture for extensible business-system tool integrations. Python / FastAPI, React 19, PostgreSQL, Redis, LDAP/OIDC.
- **Enterprise RAG & Document Vault** — Internal document management and semantic search with multi-format parsing and team-based access control. Python, FastAPI, Qdrant, Azure OpenAI, Docling, Celery, MinIO.
- **Real-Time Computer Vision for Production Lines** — YOLO inference on Raspberry Pi edge nodes, RabbitMQ + WebSocket streaming backend, React dashboard with live map. Heavy experimentation, training and deployment. PyTorch, ONNX Runtime, FastAPI, MapLibre.
- **Self-hosted OIDC Identity Server** — Rust identity foundation: OIDC, Ed25519 JWTs, device flow, TOTP MFA, LDAP. Axum, Tokio, PostgreSQL, Redis, Argon2.
- **Industrial Facial Authentication** — Distributed face-auth for the factory floor with per-operation capability authorization and full audit trail. Python + Rust, ONNX Runtime (ArcFace, YOLO-face), Qdrant, Tauri.
- **Enterprise API Gateway** — High-performance Rust gateway with routing, JWT/JWKS, rate limiting, circuit breaking and hot-reload via PostgreSQL LISTEN/NOTIFY. Axum, Hyper, Prometheus, OpenTelemetry.
- **CDC Data Integration Platform** — Manufacturing data pipeline: SQL Server Change Tracking → Avro on Kafka → ClickHouse and relational sinks. .NET 10 / C#, Rust workers, EF Core.
- **SPC Manufacturing Analytics** — Real-time SPC for SMT lines: control charts, Nelson Rules, Cp/Cpk, drift detection over CDC measurements. Python, FastAPI, SQL Server, NumPy/SciPy, React, Recharts.
- **Raspberry Pi Fleet Manager** — SSH/Ansible control plane for plant-wide Raspberry Pi fleets with artifact deployment, rollback and live progress streaming. Python, FastAPI, asyncssh, HTMX, SSE.
- **Video Capture & ML Dataset Builder** — Recording platform for Raspberry Pi camera streams with frame extraction and ZIP dataset export for CV training. FastAPI, OpenCV, WebSockets, SQL Server.
- **LLM-powered Automotive Assistant (Visteon)** — RAG-based assistant for embedded-systems documentation, code generation and troubleshooting. Python, LangChain, Azure OpenAI, FastAPI.
- **MES for Electric-Transformer Plant (Grupo Argá)** — End-to-end production / quality / traceability platform. FastAPI + PostgreSQL, Docker, Kubernetes on AWS.

SELECTED COURSES & CERTIFICATIONS

AI / ML: AI Agents, MCP, Deep Learning, Neural Networks, CNNs, AI Project Strategy, PyTorch, TensorFlow & Keras (Platzi / online)

Python & Data: Professional Python, Advanced Python, Algorithms in Python, Pandas & NumPy, Jupyter, Inferential Statistics, Stochastic Programming (Platzi)

Backend & Web: FastAPI, Flask, Django, PostgreSQL, Scrapy, Web Scraping (Platzi)

Engineering: Industrial Automation & Hydraulics — Festo; SolidWorks; EPLAN Electric; Schneider Electrical Systems

Tooling & Soft: Git & GitHub, OOP, Public Speaking, Executive Presence (Platzi)